



USER MANUAL

KRL Predictor

Application Interface Methods of Use



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1. Introduction

Welcome to KRL Predictor, a tool designed to streamline the calculation of the Local Refraction Coefficient (KRL), an essential parameter in atmospheric, geodetic, and electromagnetic wave propagation studies.

This manual will guide you step by step through the use of the software, from installation to result interpretation, ensuring that you can take full advantage of all system features.

1.1. What is the Local Refraction Coefficient (KRL)?

The Local Refraction Coefficient (KRL) is a parameter that quantifies how atmospheric conditions affect the path of electromagnetic waves as they travel through different air layers. This coefficient is fundamental for:

- Correction of high-precision geodetic measurements
- Studies on radio signal and telecommunications propagation
- Analysis of atmospheric phenomena
- Research in meteorology and climatology

1.2. Target Audience

This software is intended for:

- Scientific researchers in the fields of geodesy, geophysics, and meteorology
- Geodetic engineers performing precision measurements
- Graduate students in Earth and atmospheric sciences
- Laboratory technicians processing atmospheric data

2. System Requirements

2.1. Hardware Requirements

Component	Specification
Processor	1 GHz or higher
RAM Memory	Minimum 512 MB (2 GB recommended)
Disk Space	50 MB free
Screen Resolution	Minimum 1024 × 768 pixels

Table 1: System hardware requirements

2.2. Software Requirements

Software	Version
Operating System	Windows 10+, Linux (Ubuntu 18.04+), macOS 10.14+
Python	3.8 or higher

Table 2: Software requirements

3. User Interface

3.1. General Interface Overview

The KRL Predictor interface is designed to be intuitive and easy to use. It consists of the following elements:

- Application title: "KRL Predictor" at the top
- Mode switch: In the upper-right corner to toggle between light and dark mode
- Language switch: In the upper-right corner to toggle between Spanish and English
- Input fields: Four fields for entering atmospheric parameters
- Help buttons (?): Next to each field to display contextual information
- Action buttons: "Calculate KRL" and "Clear"
- Results area: Displays the calculated KRL value
- Visual gauge: Color-gradient bar with a position indicator

- Color legend: Interpretation of KRL ranges

3.2. Input Fields

Field	Description	Example
Altitude (m)	Measurement point altitude above sea level in metres	100
Temperature (°C)	Air temperature in degrees Celsius	20
Pressure (hPa)	Atmospheric pressure in hectopascals	1013.25
Gradient (°C/m)	Vertical temperature gradient ($\partial T/\partial Z$)	0.01

Table 3: Input field descriptions

4. Using the Software

4.1. Basic KRL Calculation

To calculate the Local Refraction Coefficient, follow these steps:

1. Launch the application by running the file KRLv_1.exe
2. Enter the measurement point altitude in metres in the first field
3. Enter the air temperature in degrees Celsius in the second field
4. Enter the atmospheric pressure in hectopascals (hPa) in the third field
5. Enter the vertical temperature gradient ($\partial T/\partial Z$) in °C/m in the fourth field
6. Click the 'Calculate KRL' button
7. Observe the result displayed below the buttons, showing the KRL value to 5 decimal places
8. Check the visual gauge to interpret the calculated KRL range

4.2. Practical Example

Scenario: Measurement at a meteorological station located 100 metres above sea level

Input data:

Parameter	Value
Altitude	100
Temperature	20
Pressure	1013.25
Gradient	0.01

Table 4: Sample input data

Expected result:

The system will calculate and display an approximate KRL value. The visual gauge will indicate whether this value falls within the low, normal, or high range.

4.3. Visual Gauge Interpretation

The visual gauge uses a colour gradient to facilitate result interpretation:

Range	KRL Values	Meaning
Low	$KRL \leq 0.11$	Stable atmospheric conditions with low refraction
Normal	$0.11 < KRL \leq 0.20$	Typical atmospheric conditions with moderate refraction
High	$KRL > 0.20$	High-refraction atmospheric conditions; significant corrections required

Table 5: Visual gauge interpretation table

4.4. Clearing Fields

To perform a new calculation with different data:

1. Click the "Clear" button
2. All input fields will be emptied automatically
3. The result and visual gauge will be reset

4. You may enter new values and calculate again

4.5. Changing Display Mode

KRL Predictor offers two display modes:

- Light Mode: White background with dark text (default)
- Dark Mode: Dark background with light text, ideal for low-light environments

To change the mode:

1. Locate the "Dark Mode" switch in the upper-right corner
2. Click the switch to toggle between modes
3. The interface will change immediately

5. Troubleshooting

5.1. Common Problems and Solutions

Problem: Error message when calculating

Symptom: "□□ Error: please enter valid numeric values"

Solutions:

- Verify that all fields contain valid numbers
- Use a period (.) as the decimal separator, not a comma (,)
- Make sure no field is left empty
- Do not include letters or special symbols in numeric values

Problem: Interface not displaying correctly

Symptom: The window appears clipped or elements are not visible

Solutions:

- Verify that your screen has a resolution of at least 1024 × 768 pixels
- Adjust the display scaling in your operating system settings
- Restart the application

5.2. Frequently Asked Questions

What units should I use?

The required units are:

- Altitude: metres (m)
- Temperature: degrees Celsius (°C)
- Pressure: hectopascals (hPa)
- Gradient: degrees Celsius per metre (°C/m)

How do I obtain the vertical temperature gradient?

The vertical temperature gradient ($\partial T/\partial Z$) is obtained by measuring the temperature difference between two known altitudes and dividing by the altitude difference. For standard atmospheric measurements, typical values range between -0.01 and 0.02 °C/m.

Are results saved automatically?

No, the current version of the software does not save results automatically. It is recommended that you note or screenshot the calculated values if you need to retain them for future reference.

Can I export the results?

Version 1.0 of KRL Predictor does not include export functionality. You may manually copy the result displayed on screen to use it in other documents or applications.

6. Glossary of Terms

Term	Definition
KRL	Local Refraction Coefficient. Parameter quantifying the effect of atmospheric refraction
Atmospheric Refraction	Deviation of electromagnetic wave paths due to variations in air density
Vertical Gradient	Rate of temperature change with respect to altitude ($\partial T/\partial Z$)
Hectopascal (hPa)	Unit of atmospheric pressure. 1 hPa = 100 pascals = 1 millibar

Geodesy	Science that studies the shape, dimensions, and gravitational field of the Earth
Tooltip	Contextual information that appears when hovering the cursor over an interface element

Table 6: Glossary